

OntoAligner-Ensemble: Voting-Based Fusion across Heterogeneous Ontology Alignment Techniques

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Abstract

Ontology alignment (OA) has evolved through several methodological paradigms, ranging from lexical and structural aligners to knowledge graph embedding (KGE) models and, more recently, Large Language Model (LLM)-based approaches. Although modern OA frameworks provide unified ecosystems for deploying these heterogeneous aligners, mechanisms for systematically reconciling their complementary and sometimes conflicting predictions remain relatively underexplored. We present OntoAligner-Ensemble, a modular and aligner-agnostic framework that combines candidate correspondences through a configurable two-stage process comprising voting-based fusion strategies (e.g., Weighted Voting, Reciprocal Rank Fusion, Condorcet, and Borda Count) followed by post-fusion selection policies. The framework supports any ontology aligner implemented within OntoAligner that produces candidate correspondences, enabling diverse alignment paradigms to be integrated through a unified decision process. To demonstrate its effectiveness, we instantiate the framework using representative lightweight string-aligner, KGE-based, and Retrieval-Augmented Generation (RAG) aligners powered by both open-weight and API-based LLMs. We evaluate individual aligners and ensemble configurations across eight benchmark tasks from five OAEI tracks spanning biomedical, material science, biodiversity, circular economy, and beyond-equivalence ontology matching. The results show that ensemble fusion improves the balance between precision and recall and frequently outperforms standalone aligners across diverse domains. Furthermore, our analysis reveals that ensemble composition directly affects the precision-recall trade-off: heterogeneous cross-paradigm ensembles generally improve precision, whereas homogeneous LLM ensembles more often achieve higher overall F1-scores. These findings demonstrate that systematic ensemble learning offers a robust and reproducible strategy for OA while providing practical guidance for selecting ensemble compositions under different alignment scenarios.

Keywords

Ontology Alignment, OntoAligner, Ensemble Learning, Voting Fusion, Large Language Models

- **URL:** <https://ontoaligner.readthedocs.io/>
- **GitHub:** <https://github.com/sciknoworg/OntoAligner>
- **PyPi:** <https://pypi.org/project/OntoAligner/>
- **License:** Apache License 2.0
- **Experimental Resources:** <https://doi.org/10.5281/zenodo.21736780>

1. Introduction

Ontology alignment (OA) is the process of finding and defining semantic correspondences between elements of different ontologies so that data and knowledge from heterogeneous sources can interoperate accurately. The origins of OA trace back to the 1980s and 1990s, with foundations in knowledge representation, artificial intelligence (AI), and database schema matching [1, 2]. With the emergence of the Semantic Web in the late 1990s and early 2000s, OA gained significant attention as widely adopted

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languages such as RDF and OWL revealed that independently developed ontologies often described similar concepts using different structures and terminologies [3, 4, 5, 6, 7]. Research accelerated rapidly following the inaugural International Semantic Web Conference (ISWC) in 2002, leading to standardized benchmarks via the Ontology Alignment Evaluation Initiative (OAEI) launched in 2004¹ and dedicated venues such as the International Workshop on Ontology Matching (OM) established in 2006² [8]. These community efforts drove advances in lexical [9], structural [10], semantic [11], and instance-based [12] techniques. Between 2016 and 2021, the field integrated machine learning and deep learning models, particularly knowledge graph embeddings (KGEs), to capture complex semantic relationships beyond surface similarities [13, 14, 15, 16]. Most recently, large language models (LLMs) have opened new frontiers in handling complex and scalable alignment processes [17, 18, 19, 20].

Although OA has historically evolved through a succession of distinct methodological paradigms often treated as competing alternatives, modern frameworks such as OntoAligner [21] aim to unify these paradigms within a modular ecosystem [22]. Yet, making heterogeneous aligners available through a common framework does not by itself determine how their predictions should be combined into a single, reliable alignment. OntoAligner enables lexical, structural, embedding-based, and generative aligners to operate through a shared pipeline, but a general and configurable mechanism for reconciling their complementary and sometimes conflicting outputs has remained missing. This exposes a central methodological question: *how can heterogeneous alignment techniques be integrated so that their strengths reinforce one another while their individual weaknesses are mitigated?* Addressing this question is particularly important because no single paradigm consistently dominates across ontology domains. Lexical matchers can achieve high recall when ontologies are terminologically similar but may degrade under semantic drift; KGE-based aligners can effectively capture relational structures but remain sensitive to graph sparsity; and LLM-driven aligners can generalize well in open-vocabulary contexts, although their performance varies considerably across models, providers, and domains.

Rather than treating this variability solely as a limitation, we view it as motivation for systematic ensemble alignment. Contemporary OA toolkits increasingly make heterogeneous alignment techniques interoperable, yet providing them within a common framework does not by itself determine how their potentially complementary and conflicting predictions should be combined. Although recent work has introduced retrieval-augmented generation (RAG) [17, 19] and in-context reasoning into OA, LLM-based aligners have largely been studied individually, while prior work within OntoAligner [22] has primarily considered open-weight models. Consequently, a unified and configurable mechanism for combining conventional, KGE-based, and LLM-driven aligners, including both open-weight and API-based models, remains insufficiently investigated. To address this gap, this work makes the following central contribution:

We introduce OntoAligner-Ensemble, a unified and configurable ensemble mechanism that combines voting-based fusion with post-fusion selection to integrate predictions from heterogeneous ontology aligners, and evaluate it across multiple domains using lightweight, KGE-based, and RAG-based aligners built on both open-weight and API-based LLMs.

We evaluate the contribution through three research questions: **RQ1:** How do lightweight, KGE-based, and RAG/LLM-based aligners compare across heterogeneous OA tasks in terms of precision, recall, and F1-score? **RQ2:** To what extent can voting-based ensemble fusion improve alignment quality and balance precision and recall relative to individual aligners and established task baselines? **RQ3:** How does ensemble composition affect performance, particularly when comparing a heterogeneous ensemble spanning multiple alignment paradigms with a homogeneous ensemble of LLM-based aligners? To investigate these questions, we evaluate five aligners and two ensemble configurations across eight tasks from five OAEI tracks spanning beyond-equivalence, circular-economy, anatomy, materials-science, and biodiversity settings. The evaluation compares conventional similarity-based, KGE-based, and RAG/LLM-based approaches, including models from open-weight and API-based ecosystems, under OA scenarios that differ in domain, scale, semantic complexity, and structural characteristics.

¹<https://oaei.ontologymatching.org/>

²<http://om2006.ontologymatching.org/>

The remainder of this paper is organized as follows: Section 2 formalizes the OA task; Section 3 reviews related work; Section 4 presents the proposed ensemble alignment framework; Section 5 describes the experimental setup and evaluation methodology; Section 6 discusses the empirical findings; and Section 7 concludes the paper and outlines future research directions.

2. Problem Formalization

Let $C_{\text{source}} \in O_{\text{source}}$ and $C_{\text{target}} \in O_{\text{target}}$ denote concepts from the source ontology O_{source} and the target ontology O_{target} , respectively. Formally, the OA is defined as a:

$$OA(O_{\text{source}}, O_{\text{target}}) := M \quad (1)$$

where

$$M = \{(s, t, r, S_r) \mid s \in O_{\text{source}}, t \in O_{\text{target}}\} \quad (2)$$

and $r \in \{\equiv, \sqsubseteq, \sqsupseteq\}$ denotes the semantic correspondence between the source concept s and the target concept t , representing *equivalence*, *subsumption* (subclass), or *inverse subsumption* (superclass), respectively. The confidence score $S_r \in [0, 1]$ indicates the likelihood of the predicted correspondence.

3. Related Work

OA has traditionally relied on combining lexical, structural, and semantic similarity measures using manually designed aggregation strategies [23]. While these approaches have shown competitive performance, their effectiveness often depends on the characteristics of the ontologies being aligned, motivating the adoption of machine learning techniques to learn better combinations of matching evidence automatically. One of the earliest machine learning approaches was proposed by Eckert et al. [24], who introduced a meta-level learning framework that combines the outputs of multiple ontology aligners using a supervised classifier. Their results demonstrated that learned ensembles consistently outperform individual matchers and simple voting strategies. Later work of Nkisi-Orji et al. [25] has extended this idea by incorporating richer semantic representations. For example, Random Forest-based OA combines traditional similarity measures with word embedding features, enabling the model to learn effective combinations without manually tuning similarity weights.

Several studies have further explored ensemble learning for OA. The ROME framework [26] applies Bagging and Boosting techniques to improve robustness and generalization across OA tasks. Similarly, Xue et al. [27] introduced a collaborative OA based on dual-population genetic programming that employs active meta-learning and a Random Forest meta-classifier to construct high-quality similarity features and improve alignment accuracy. Interactive approaches such as DualLoop [28] also adopt ensemble principles by combining multiple heuristic matchers with active learning to reduce human annotation effort.

Ensemble techniques have also been applied in domain-specific settings. For instance, biomedical OA has benefited from integrating word embedding similarities with existing ontology aligners [29], demonstrating that learned semantic features complement traditional matching methods. Although focusing on instance matching, [30] introduced a hybrid ensemble classifier for identifying equivalent entities across heterogeneous datasets, providing insights that are also applicable to ontology matching and semantic data integration.

Overall, these studies demonstrate that ensemble learning can enhance OA by combining complementary similarity measures, matching strategies, and alignment paradigms. However, ensemble-based approaches remain relatively underexplored within the broader history of OA research. A possible explanation is the additional computational overhead introduced by combining multiple aligners, which can increase execution time and resource requirements. Although this trade-off may limit their applicability in time-sensitive scenarios, it can be justified when accuracy, robustness, and the ability to handle heterogeneous ontologies are prioritized over response efficiency. These observations motivate further

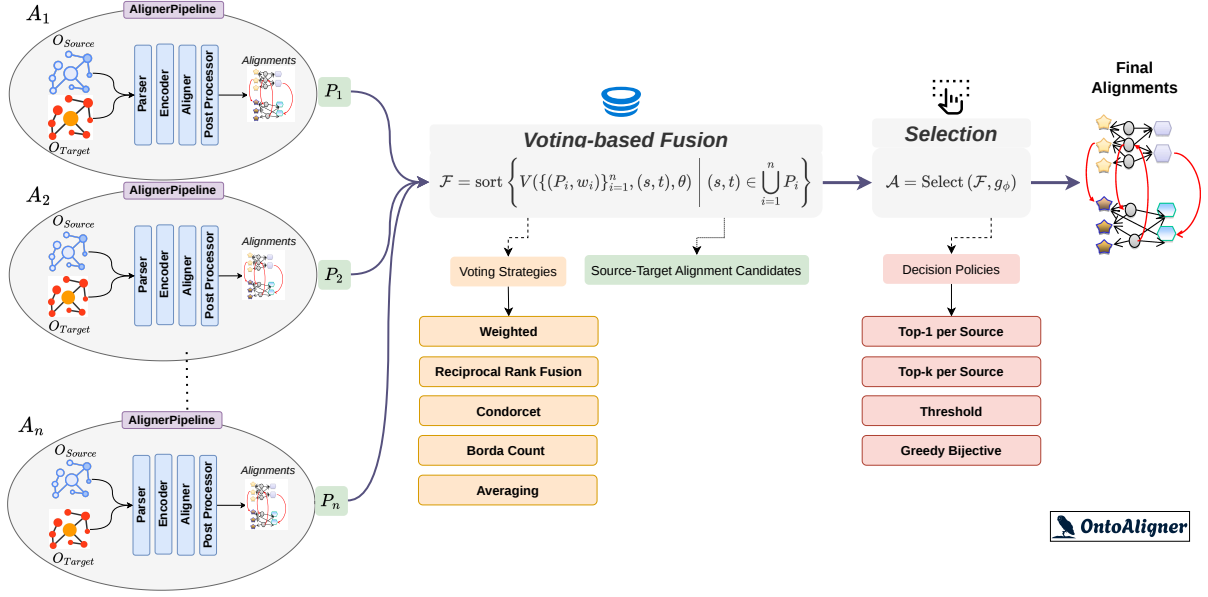


Figure 1: Overview of the OntoAligner-Ensemble architecture. Individual aligner pipelines ($A_1 \dots A_n$) process source (O_{Source}) and target (O_{Target}) ontologies through parsing, encoding, alignment, and post-processing steps. Candidate alignments (P_i) are combined using a configurable Voting-based Fusion layer—employing one of the strategies such as Weighted, Reciprocal Rank Fusion, Condorcet, Borda Count, or Averaging. Final alignments ($\mathcal{A}$) are generated after applying candidate selection and decision policies (e.g., Top-1, Top- k , Threshold, or Greedy Bijective).

investigation of ensemble strategies that can combine the complementary strengths of diverse ontology alignment methods while maintaining a flexible and practical architecture.

OntoAligner-Ensemble addresses this gap primarily through its scope and integration strategy rather than through a fundamentally new aggregation mechanism. Eckert et al. [24] and Nkisi-Orji et al. [25] combine matcher outputs using supervised meta-classifiers, requiring labeled training data for the meta-learner. ROME [26] applies Bagging and Boosting within a single learning paradigm, while Xue et al. [27] employ genetic programming and active meta-learning to construct similarity features. In contrast, OntoAligner-Ensemble provides training-free, configurable voting-based fusion across aligners from fundamentally different methodological paradigms, including lexical, KGE-based, and RAG/LLM-based approaches. It further supports both open-weight and API-based LLMs, which are integrated through a shared `AlignerPipeline` interface. This design enables heterogeneous aligners to be combined within a common ensemble framework without requiring a separately trained meta-learner.

4. OntoAligner-Ensemble Framework

Due to the complexity and diversity of real-world ontological structures, individual aligner approaches might capture only specific aspects of semantic similarity, such as lexical information, structural characteristics, or contextual knowledge. To address this limitation, we propose the OntoAligner-Ensemble framework that combines multiple independent alignment pipelines through a configurable voting-based aggregation mechanism. The proposed framework is illustrated in Figure 1 and follows a two-stage decision process: 1) *Fusion stage*, multiple aligners generate candidate correspondences, which are aggregated using a voting strategy to obtain a unified ranking of candidate alignments. 2) *Selection stage*, a post-fusion selection strategy converts the ranked candidate space into the final alignment by applying task-specific constraints.

The framework is designed to be independent of the underlying alignment algorithms. Any ontology aligner capable of producing source–target correspondence predictions can be incorporated as a constituent aligner; all these capabilities are possible because of the high-level abstraction of

AlignerPipeline module³ that is responsible for running the aligner steps, including parsing, encoding, aligner, and post-processing (if necessary), all of which are according to the OntoAligner backbone architecture. In the following, we describe the principal components of OntoAligner-Ensemble.

Ensemble Representation. Let an ensemble consist of n independent aligners $A = \{A_1, A_2, \dots, A_n\}$, where each aligner A_i corresponds to an individual AlignerPipeline. Each aligner produces a set of candidate correspondences: $P_i = \{(s, t, q_i(s, t))\}$, where $s \in O_{source}$, $t \in O_{target}$ and $q_i(s, t)$ is the confidence score that A_i assigns to the candidate correspondence (s, t) . To account for differences in reliability among aligners, each aligner is assigned a weight $w_i \in \mathbb{R}_{\geq 0}$ (the default is set to 1), where larger values indicate a greater contribution during the fusion process. Therefore, the complete ensemble input is represented as:

$$\mathcal{P} = \{(P_i, w_i)\}_{i=1}^n \quad (3)$$

Once the predictions have been generated by the n aligners, the framework converts them into a unified representation because different aligners may use different output formats. For example, the output of a retrieval-based aligner is converted from $s \mapsto \{(t_1, q_1), (t_2, q_2), \dots\}$ to individual triples (s, t_j, q_j) . This allows different aligners to participate in the same ensemble process. Additionally, duplicate (s, t) are removed by retaining the highest-ranked occurrence according to the confidence score: $(s, t, q) = \operatorname{argmax}_q(s, t)$. This guarantees a unique candidate representation before voting.

Voting-Based Fusion. The fusion component aggregates predictions from multiple constituent aligners. Let

$$\mathcal{C} = \bigcup_{i=1}^n \{(s, t) \mid (s, t, q_i(s, t)) \in P_i\} \quad (4)$$

denote the set of candidate source–target pairs proposed by at least one aligner. For each candidate $(s, t) \in \mathcal{C}$, the fused score is defined as

$$S(s, t) = \mathcal{V}(\{(P_i, w_i)\}_{i=1}^n, (s, t), \theta) \quad (5)$$

where $\mathcal{V}$ denotes the selected voting strategy and θ contains its strategy-specific parameters. The ranked fused prediction set is then defined as

$$\mathcal{F} = \operatorname{sort}_{\downarrow S}(\{(s, t, S(s, t)) \mid (s, t) \in \mathcal{C}\}) \quad (6)$$

where candidates are ordered by decreasing fused score. OntoAligner-Ensemble supports weighted voting, Reciprocal Rank Fusion (RRF), Condorcet voting, Borda count, and score averaging.

Post-Fusion Selection. After fusion, the framework contains a ranked candidate pool $\mathcal{F}$. A selection operator converts this candidate pool into the final alignment of $\mathcal{A} = \operatorname{Select}(\mathcal{F}, g_\phi)$, where ϕ represents the selection strategy parameters. Moreover, g_ϕ is defined as a decision policy that is responsible for selecting appropriate alignments based on its own selection policy; the default for g_ϕ is none. However, there are four different decision policies which are defined as follows:

- **Top-1 per Source Selection.** This decision policy aims to select the highest scoring target for each source: $t^* = \operatorname{argmax}_t S(s, t)$.
- **Top- k per Source Selection.** For each source entity, the framework retains the top- k highest-scoring target candidates. Using this policy, optionally, candidates are first filtered using a relative score margin $S(s, t) \geq m \cdot \max_{t'} S(s, t')$, where $m \in [0, 1]$ is the margin parameter. When both constraints are specified, only the top- k candidates satisfying the margin criterion are retained.
- **Threshold-based Selection.** Only correspondences exceeding a confidence threshold are retained: $\mathcal{A} = \{(s, t) \in \mathcal{F} \mid S(s, t) \geq \gamma\}$, where γ is the confidence threshold.

³See AlignerPipeline usage at <https://ontoaligner.readthedocs.io/developerguide/pipeline.html>

Table 1

OAEI tracks and tasks statistics across source, target, and alignments.

Track	Task	O_{source} Classes	O_{target} Classes	References
<i>Beyond Equivalence</i>	G1 - Web	727	1,132	339
	G2 - Diseases	1,108	5,145	355
	G3 - Text	334	259	762
<i>Circular Economy</i>	CEON - BiOnto	228	779	29
	CEON - MatOnto	228	846	16
<i>Anatomy</i>	Mouse-Human	2,743	3,304	1,516
<i>Material Science and Engineering</i>	MI-MatOnto	545	847	302
<i>Biodiversity and Ecology</i>	Fish-Zooplankton	145	56	15

- **Greedy Bijective Selection.** For applications requiring one-to-one alignment, this decision policy applies a greedy bijective selection strategy as follows: $\forall (s_i, t_i), (s_j, t_j) \in \mathcal{A}$, where $s_i \neq s_j \wedge t_i \neq t_j$, meaning that each source and target entity can appear at most once.

This modular formulation enables OntoAligner-Ensemble to integrate heterogeneous OA systems within a unified framework while supporting configurable voting strategies and post-fusion selection policies that can be adapted to different datasets and application requirements ⁴.

5. Experimental Setup

Evaluation Datasets: OAEI Tracks and Tasks. We selected eight OA tasks from five tracks of the OAEI campaign to evaluate OntoAligner-Ensemble across diverse domains. The statistics of the selected datasets, including the number of source ontology classes, target ontology classes, and reference correspondences, are summarized in Table 1. The selected tracks include: *Beyond Equivalence* [31, 32] (G1-Web, G2-Diseases, and G3-Text), *Circular Economy* [33] (CEON-BiOnto and CEON-MatOnto), *Anatomy* [34] (Mouse-Human), *Material Science and Engineering* [35] (MI-MatOnto), and *Biodiversity and Ecology* [36] (Fish-Zooplankton). These datasets were selected to provide a comprehensive evaluation across heterogeneous domains, ontology sizes, and matching difficulties.

The *Beyond Equivalence* track evaluates OA systems beyond simple equivalence detection by considering diverse semantic relations, including equivalence, subsumption, overlap, and disjointness. It contains heterogeneous benchmarks from industrial classification schemes and the STROMA/TaSeR repository. In this work, we use the STROMA/TaSeR-based tasks (G1-Web, G2-Diseases, and G3-Text), which provide diverse matching scenarios with varying ontology sizes and semantic complexity. The *Circular Economy* and *Material Science and Engineering* tracks evaluate domain-specific ontologies with relatively smaller alignment spaces, while the *Anatomy* track represents a larger matching task with thousands of ontology classes and reference correspondences. Finally, the *Biodiversity and Ecology* task provides a small-scale evaluation scenario with limited reference alignments. Overall, the selected benchmarks cover a broad range of task characteristics, including small-scale tasks (Fish-Zooplankton, CEON-BiOnto, and CEON-MatOnto), medium-scale tasks (G1-Web, G3-Text, and MI-MatOnto), and large-scale tasks (G2-Diseases and Mouse-Human). This diversity enables a comprehensive assessment of the robustness and generalization capability of both individual OA systems and the proposed ensemble framework.

Individual Aligners. To evaluate both individual OA systems and their ensemble combinations, we selected five aligners representing three complementary categories: *Lightweight*, *KGE* [22], and *RAG* [19]. All aligners were implemented using the `AlignerPipeline` interface provided by OntoAligner, enabling a unified execution and prediction format across heterogeneous alignment approaches.

- *Lightweight.* We employed the fuzzy string matching aligner available in OntoAligner⁵, using a

⁴See `EnsembleLearningAligner` usage at https://ontoaligner.readthedocs.io/aligner/ensemble_learning.html

⁵<https://ontoaligner.readthedocs.io/aligner/lightweight.html>

similarity threshold of 0.7 for all experiments.

- *KGE*. We selected the ConvE-based aligner with the same hyperparameter configuration reported in our previous work [22].
- *RAG*. To evaluate the robustness of the proposed ensemble across different LLM ecosystems, we considered three RAG configurations:
 - *Qwen*. Qwen3.5-9B [37] as the generator and Qwen3-Embedding-4B [38] as the retriever, both from the Qwen model family.
 - *GPT*. GPT-5.4-Nano⁶ as the generator and text-embedding-3-small⁷ as the retriever, representing the OpenAI model family.
 - *Gemini*. The Gemini 2.5 Flash-Lite⁸ as the generator and EmbeddingGemma-300m [39] as the retriever, representing the Google model family.

The hyperparameter values used throughout the experiments were selected manually based on preliminary empirical observations and prior experience with the underlying alignment methods. No systematic hyperparameter optimization was performed, as the large configuration space arising from multiple aligners, voting strategies, and selection policies would make exhaustive or automated tuning computationally prohibitive.

Ensemble Configurations. To evaluate the effectiveness of ensemble-based aligners, we constructed two configurations of OntoAligner-Ensemble. Each ensemble treats individual aligners as independent techniques and combines their predictions through a voting-based fusion strategy followed by a decision policy.

- *Ens. (All)*. This configuration combines all available aligners across different categories, including the lightweight string-based aligner, the KGE-based aligner, and the three RAG-based aligners. This configuration evaluates whether heterogeneous alignment systems with complementary characteristics can improve overall alignment performance through collective decision-making.
- *Ens. (LLMs)*. This configuration focuses exclusively on LLM-based aligners and combines the three RAG aligners, namely Qwen, GPT, and Gemini. This setting investigates whether an ensemble of different LLM-based aligners can provide more robust predictions compared with individual LLM aligners.

For all ensemble experiments, each aligner was assigned an equal weight ($w_i = 1.0$). The majority voting strategy (supported by weighted voting) was employed with a minimum valid vote of 3 for *Ens. (All)* and 2 for *Ens. (LLMs)*. After fusion, the resulting ranked correspondence candidates were processed using the *top-1-source* decision policy to obtain the final alignment set. This configuration was selected to provide a consistent comparison across datasets while avoiding additional tuning of ensemble-specific parameters. Nevertheless, the proposed framework is fully configurable, allowing practitioners to adjust both model-specific and ensemble-level hyperparameters to suit different datasets, computational budgets, and application requirements.

Baseline Provenance. The baseline results reported in Table 2 for MDMapper, LogMapBio, LogMap, LogMapLt, and Matcha are taken directly from official OAEI campaign results.

6. Results and Discussion

We evaluate the five individual aligners and the two proposed ensemble configurations, *Ens. (All)* and *Ens. (LLMs)*, across eight benchmark tasks from five OAEI tracks. The experimental results, measured

⁶<https://developers.openai.com/api/docs/models/gpt-5.4-nano>

⁷<https://developers.openai.com/api/docs/models/text-embedding-3-small>

⁸<https://ai.google.dev/gemini-api/docs/models/gemini-2.5-flash-lite>

Table 2

Ontology matching results across all evaluation datasets. “*Base.*” denotes the baseline aligner for each dataset. Best precision, recall, and F1-score for each task are shown in bold.

Task	Base. Aligner	Metric	Aligners							
			Base.	Fuzzy	KGE	Qwen	GPT-5.4	Gemini	Ens. (All)	Ens. (LLMs)
G1-Web	MDMapper [40]	Precision	88.2	39.8	68.9	65.5	56.3	61.7	74.2	68.7
		Recall	36.3	53.0	41.2	51.0	54.8	51.9	48.3	51.3
		F1	51.5	45.5	51.6	57.3	55.6	56.4	58.5	58.7
G2-Diseases	LogMapBio [10]	Precision	60.1	32.5	60.1	50.1	48.6	51.2	55.2	54.6
		Recall	61.0	69.2	54.3	72.9	75.7	71.5	69.4	74.9
		F1	60.5	44.3	57.1	59.4	59.2	59.6	61.4	63.1
G3-Text	LogMap [10]	Precision	43.7	40.0	36.7	48.3	41.7	45.2	45.9	46.9
		Recall	7.3	9.4	5.1	9.0	8.9	8.7	8.13	9.0
		F1	12.5	15.2	8.9	15.0	14.7	14.7	13.8	15.2
CEON-BiOnto	LogMap [10]	Precision	67.6	32.9	78.2	47.4	40.2	51.7	63.6	55.7
		Recall	86.2	93.1	62.0	96.5	100	100	96.5	100
		F1	75.8	48.6	69.2	63.6	57.4	68.2	76.7	71.6
CEON-MatOnto	LogMap [10]	Precision	35.6	22.8	33.3	27.7	27.7	28.0	33.3	29.4
		Recall	100	100	81.2	93.7	93.7	100	93.7	93.7
		F1	52.5	37.2	47.2	42.8	42.8	43.8	49.1	44.7
Fish-Zooplankton	LogMapLt [10]	Precision	80.0	64.2	100	92.8	93.3	100	100	100
		Recall	53.3	60.0	53.3	86.6	93.3	86.6	73.3	93.3
		F1	64.0	62.0	69.5	89.6	93.3	92.8	84.6	96.5
MI-MatOnto	Matcha [41]	Precision	75.6	39.3	93.3	61.7	55.0	63.5	84.3	72.7
		Recall	21.9	21.5	9.2	32.1	27.1	29.4	23.1	29.1
		F1	33.9	27.8	16.8	42.2	36.3	40.2	36.3	41.6
Mouse-Human	Matcha [41]	Precision	95.1	53.8	99.0	86.9	87.0	89.4	96.5	93.0
		Recall	93.1	76.7	61.3	90.6	89.3	88.3	85.5	90.3
		F1	94.1	63.2	75.6	88.7	88.1	88.8	90.6	91.6

in terms of precision, recall, and F1-score, are summarized in Table 2. The remainder of this section is organized around the three research questions: RQ1 examines the performance profiles of the individual aligners, RQ2 evaluates the effectiveness of the proposed voting-based ensemble, and RQ3 investigates how ensemble composition affects performance across domains and task types.

6.1. RQ1: Performance of Individual Aligners

No individual aligner consistently achieves the best performance across all evaluation tasks. Instead, the strongest approach varies with the benchmark, indicating that OA remains task-dependent. The three evaluated aligner categories exhibit distinct precision-recall profiles.

- **Lightweight.** The fuzzy string aligner achieves high recall on tasks with substantial lexical overlap, including *CEON-BiOnto* (93.1%) and *CEON-MatOnto* (100.0%). However, this behavior is accompanied by comparatively low precision, such as 32.5% on *G2-Diseases* and 22.8% on *CEON-MatOnto*. These results indicate that reliance on surface-form similarity can retrieve many candidate correspondences while also introducing false positives.
- **KGE-based.** The ConvE-based KGE aligner achieves high precision on several tasks, consistent with observations in prior work [22]. It reaches 100.0% precision on *Fish-Zooplankton* and 93.3% on *MI-MatOnto*. This precision-oriented behavior is nevertheless accompanied by low recall on some tasks, including 9.2% on *MI-MatOnto* and 5.1% on *G3-Text*.
- **RAG/LLM-based.** The RAG-based aligners generally provide broader correspondence coverage, although their performance varies across models and tasks. On *MI-MatOnto*, Qwen achieves an F1-score of 42.2%, outperforming the established baseline and all other individual aligners. On *Fish-Zooplankton*, GPT-5.4-Nano achieves a balanced precision and recall of 93.3%. The three LLM-based aligners nevertheless exhibit different performance profiles; for example, the precision of GPT-5.4-Nano ranges from 40.2% on *CEON-BiOnto* to 93.3% on *Fish-Zooplankton*. Thus, no individual LLM is consistently superior across all tasks.

For six of the eight tasks, all three RAG/LLM-based aligners achieve higher recall than the corresponding established baseline. The exceptions are *CEON-MatOnto*, where the baseline already achieves 100.0% recall, and *Mouse-Human*, where the baseline recall of 93.1% exceeds that of the three RAG aligners. The recall improvement is particularly pronounced on *Fish-Zooplankton*, where GPT-5.4-Nano reaches 93.3% recall compared with 53.3% for LogMapLt.

The broader coverage of the RAG/LLM-based aligners is often accompanied by lower precision than that of the established baselines. Conversely, the KGE aligner frequently achieves high precision but lower recall. The lightweight aligner can also obtain high recall in lexically similar tasks, although often with a substantial precision reduction. These complementary performance profiles provide the motivation for the ensemble analysis in RQ2.

Established task baselines nevertheless remain competitive. They achieve the highest F1-score on *CEON-MatOnto* and *Mouse-Human* and remain close to the best-performing approach on tasks such as *G2-Diseases* and *CEON-BiOnto*. Overall, RQ1 shows that the relative effectiveness of lightweight, KGE-based, and RAG/LLM-based aligners depends on the task: lightweight and RAG-based approaches generally favor recall, whereas the evaluated KGE aligner generally favors precision, and no individual alignment paradigm dominates across all benchmarks.

6.2. RQ2: Effectiveness of the Proposed Voting-Based Ensemble

The results indicate that voting-based fusion can improve the balance between precision and recall, although its effectiveness remains task-dependent. At least one ensemble configuration achieves or ties for the highest F1-score on five of the eight tasks: *G1-Web*, *G2-Diseases*, *G3-Text*, *CEON-BiOnto*, and *Fish-Zooplankton*. These results show that combining predictions from multiple aligners can improve upon the constituent systems when their predictions provide complementary alignment evidence.

The clearest improvements are observed on *G1-Web*, *G2-Diseases*, and *Fish-Zooplankton*. On *G1-Web*, *Ens. (LLMs)* achieves the highest F1-score of 58.78%, exceeding the best individual aligner, Qwen, at 57.37%, and the MDMapper baseline at 51.50%. On *G2-Diseases*, *Ens. (LLMs)* reaches 63.1% F1, compared with 59.6% for the strongest individual aligner, Gemini, and 60.5% for the LogMapBio baseline. The largest ensemble gain occurs on *Fish-Zooplankton*, where *Ens. (LLMs)* obtains 96.5% F1, improving upon the strongest individual aligner, GPT-5.4-Nano, at 93.3%, and the LogMapLt baseline at 64.0%.

The ensemble configurations also provide competitive results on *G3-Text* and *CEON-BiOnto*. On *G3-Text*, *Ens. (LLMs)* reaches 15.2% F1, tying the fuzzy string aligner for the highest result and exceeding the LogMap baseline at 12.5%. On *CEON-BiOnto*, *Ens. (All)* achieves the highest overall F1-score of 76.7%, slightly surpassing the LogMap baseline at 75.8% and outperforming all individual aligners.

However, voting-based fusion does not uniformly outperform the strongest individual method or established baseline. On *MI-MatOnto*, Qwen achieves the highest F1-score of 42.2%, compared with 41.6% for *Ens. (LLMs)* and 36.3% for *Ens. (All)*. On *CEON-MatOnto*, the LogMap baseline remains strongest at 52.5% F1, whereas *Ens. (All)* and *Ens. (LLMs)* achieve 49.1% and 44.7%, respectively. Similarly, on *Mouse-Human*, the Matcha baseline retains the highest F1-score of 94.1%, followed by *Ens. (LLMs)* at 91.6% and *Ens. (All)* at 90.6%.

The results therefore show that voting-based fusion is most effective when agreement among the constituent aligners suppresses individual errors without removing too many valid correspondences. When one constituent aligner is already substantially stronger than the others, or when voting reduces recall too aggressively, the ensemble may remain below the strongest individual method or specialized task baseline.

Ensembling also introduces an expected computational trade-off because the predictions of multiple constituent aligners must be generated and combined. Runtime was not evaluated as an experimental metric in this study, so no quantitative efficiency comparison can be made. Nevertheless, the modular implementation within OntoAligner allows the number and types of constituent aligners to be adjusted according to task requirements and the available computational budget. Overall, RQ2 shows that the proposed voting-based ensemble can improve alignment quality and the precision-recall balance, but its benefits depend on the benchmark and the relative strengths of its constituent aligners.

6.3. RQ3: Effect of Ensemble Composition across Domains

The two ensemble configurations exhibit distinct precision–recall profiles. The heterogeneous configuration, *Ens. (All)*, combines the lightweight, KGE-based, and three RAG/LLM-based aligners, whereas the homogeneous configuration, *Ens. (LLMs)*, combines only the three RAG/LLM-based aligners. Across the eight tasks, *Ens. (All)* generally favors precision, while *Ens. (LLMs)* more often preserves recall. Consequently, *Ens. (LLMs)* achieves a higher F1-score than *Ens. (All)* on six tasks, whereas *Ens. (All)* performs better on the two Circular Economy tasks.

- **Homogeneous LLM ensembling (*Ens. (LLMs)*).** Combining Qwen, GPT-5.4-Nano, and Gemini through majority voting achieves or ties for the highest F1-score on four tasks: *G1–Web* (58.78%), *G2–Diseases* (63.1%), *G3–Text* (15.2%), and *Fish–Zooplankton* (96.5%). The configuration outperforms every standalone LLM on *G1–Web*, *G2–Diseases*, and *Fish–Zooplankton*. Its comparatively strong recall suggests that agreement among the LLM-based aligners retains broader semantic coverage than the more heterogeneous ensemble.
- **Heterogeneous multi-paradigm ensembling (*Ens. (All)*).** Combining lightweight, KGE-based, and RAG/LLM-based aligners produces a more precision-oriented configuration. On *Mouse–Human*, *Ens. (All)* achieves 96.5% precision, exceeding all three individual LLM aligners, although the KGE aligner remains higher at 99.0%. On *MI–MatOnto*, *Ens. (All)* reaches 84.3% precision, which is 22.6 percentage points higher than the best individual LLM precision of 61.7% obtained by Qwen. Its recall on this task is only 23.1%, however, resulting in an F1-score of 36.3%. The heterogeneous ensemble therefore acts as a comparatively conservative filter: agreement across different alignment paradigms can increase precision, but may also remove valid correspondences that are supported by only a subset of the constituent aligners.

The effect of ensemble composition also differs across domains and task types:

- **Biomedical and standardized domains (*Mouse–Human*, *G2–Diseases*).** On *Mouse–Human*, the specialized Matcha baseline retains the highest F1-score of 94.1%. Nevertheless, *Ens. (LLMs)* achieves a competitive 91.6% F1 in the reported setup without task-specific fine-tuning, while *Ens. (All)* reaches 90.6%. The heterogeneous ensemble obtains higher precision than the LLM ensemble (96.5% versus 93.0%), whereas the LLM ensemble preserves higher recall (90.3% versus 85.5%). On *G2–Diseases*, *Ens. (LLMs)* achieves 63.1% F1 and outperforms both *Ens. (All)* at 61.4% and the LogMapBio baseline at 60.5%. These results indicate that the LLM-only composition can provide competitive coverage of biomedical terminology, although the advantage is not uniform across biomedical tasks.
- **Materials science and Circular Economy (*MI–MatOnto*, *CEON–BiOnto*, *CEON–MatOnto*).** On *MI–MatOnto*, Qwen achieves the highest F1-score of 42.2%, followed closely by *Ens. (LLMs)* at 41.6%, both exceeding the Matcha baseline at 33.9% and the KGE aligner at 16.8%. In contrast, *Ens. (All)* achieves substantially higher precision (84.3%) but lower recall (23.1%), resulting in 36.3% F1. This illustrates how the heterogeneous composition can favor conservative correspondences at the expense of broader coverage. On *CEON–BiOnto*, *Ens. (All)* achieves the highest F1-score of 76.7%, exceeding the LogMap baseline at 75.8% and *Ens. (LLMs)* at 71.6%. The LLM ensemble reaches 100.0% recall, but its lower precision of 55.7% reduces its overall F1-score. On *CEON–MatOnto*, neither ensemble exceeds the LogMap baseline at 52.5% F1. Nevertheless, *Ens. (All)* again outperforms *Ens. (LLMs)* in F1-score (49.1% versus 44.7%) because of its higher precision (33.3% versus 29.4%), while both configurations obtain the same recall of 93.7%. The two CEON tasks therefore provide the clearest cases in which cross-paradigm agreement improves the precision–recall balance relative to the LLM-only composition.
- **Non-equivalence and complex mapping (*G3–Text*).** Recall remains below 10% for every evaluated method on *G3–Text*, indicating that this task remains challenging for both individual and ensemble approaches. In this setting, *Ens. (LLMs)* achieves 15.2% F1, tying the fuzzy string

aligner for the highest result and exceeding both *Ens. (All)* at 13.8% and the LogMap baseline at 12.5%. The LLM-only composition therefore provides a modest advantage over the heterogeneous ensemble under this difficult mapping setting, although the low absolute recall indicates substantial remaining room for improvement.

- **Biodiversity and ecology (*Fish–Zooplankton*).** Both ensemble configurations achieve 100.0% precision, but their recall differs substantially. *Ens. (LLMs)* reaches 93.3% recall and 96.5% F1, whereas *Ens. (All)* obtains 73.3% recall and 84.6% F1. The lower recall of the heterogeneous ensemble suggests that requiring agreement across substantially different alignment paradigms can be overly restrictive for this task. Here, the homogeneous LLM composition retains more valid correspondences while preserving perfect precision.

Overall, RQ3 shows that ensemble composition directly affects the precision–recall trade-off. The heterogeneous multi-paradigm ensemble generally provides stronger precision and performs best on the two Circular Economy tasks, whereas the homogeneous LLM ensemble generally retains more recall and achieves the higher F1-score on six of the eight tasks. Neither composition is uniformly preferable; their relative effectiveness depends on the domain, the task characteristics, and the degree of agreement among the constituent aligners.

7. Discussion and Conclusion

Industry Perspective. From an industrial application perspective, scaling the alignment and integration of proprietary and domain-standard ontologies directly supports the deployment of reliable data products for both human analysts and automated agents. Achieving robust semantic interoperability is crucial in regulatory reporting, reference data integration, and enterprise analytics, where data assets created for distinct business purposes must be harmonized into unified semantic layers. In these settings, raw individual aligners often face operational trade-offs: lexical aligners can fail under proprietary terminology changes, while LLM-driven aligners introduce variance across model providers. The OntoAligner-Ensemble mechanisms developed in this work help mitigate these single-model risks by providing predictable, high-precision candidate alignments. Improving the precision–recall balance through systematic fusion lowers the manual verification cost needed to build trusted data products, ultimately facilitating smoother data consumption, discovery, and inter-organizational system integration.

Limitations and Future Directions. While our OA definition considers four matching relations (*equivalence*, *subclass*, *superclass*, and *disjointness*), the current framework reduces output decisions to binary classification (*match* vs. *no-match*). Consequently, while a positive *match* can implicitly encompass equivalence, subclass, and superclass associations, the framework does not differentiate between these distinct relation types. This limitation stems from the current fusion functions, which prioritize consensus over fine-grained semantic categorization. Extending the decision layer to distinguish specific semantic relations represents a key direction for future developments of OntoAligner. Moreover, several evaluated tasks have small reference alignments (*Fish–Zooplankton*: 15 correspondences; *CEON–MatOnto*: 16; *CEON–BiOnto*: 29), so reported percentage-point differences on these tasks can correspond to only one or two changed predictions and should be read as indicative rather than statistically robust. The reported ensemble results also instantiate a single fusion strategy and selection policy out of those supported by the framework; evaluating the remaining configurations, and incorporating formal-semantics-based fusion alongside the statistical voting strategies studied here, are promising directions for future work.

Conclusion. In this work, we presented OntoAligner-Ensemble, a modular, aligner-agnostic framework designed to systematically integrate complementary candidate predictions across heterogeneous OA paradigms. By employing a two-stage pipeline combining voting-based fusion strategies with

post-fusion selection policies, the framework reconciles conflicting correspondences generated by diverse aligner algorithms. We systematically evaluated standalone and ensemble aligners across eight benchmark tasks from five OAEI tracks. Our experimental findings demonstrate that voting-based ensemble fusion improves the precision–recall balance, frequently outperforming individual constituent baselines across varying domain complexities.

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Declaration on Generative AI

In preparing this manuscript, generative AI tools, specifically ChatGPT, were used solely for grammar checking, spelling checks, and readability of some sentences. All suggested changes were carefully reviewed and adapted by the authors to ensure accuracy and appropriateness. The scientific content, research design, analysis, and conclusions were developed and verified exclusively by the authors without AI involvement. The use of ChatGPT was limited to enhancing the presentation of the work. Generative AI (GenAI) tools were used solely to assist with code development for conducting the experiments reported in this paper

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